H524 Introduction to Biostatistics

Week 2: Probability Concepts and Applications

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Outline

Quick Review: Key Concepts

- **Data Types:** Quantitative vs Qualitative
- **Data Presentation:** Histograms, box plots, bar charts
- **Summary Measures:** Central tendency and variability
- **Rates and Standardization:** Age-specific rates, direct/indirect methods

This Week: Chapter 6 - Probability foundations for statistical inference

Reading: Pagano & Gauvreau Chapter 6 (6.1-6.4.2, 6.5)

Why Probability in Biostatistics?

- **Foundation for Statistical Inference**
  - Understanding uncertainty in medical research
  - Bridge between sample data and population conclusions
- **Clinical Applications**
  - Diagnostic test interpretation
  - Treatment efficacy assessment
  - Risk communication to patients
- **Research Design**
  - Power calculations
  - Sample size determination
  - Study planning and interpretation

Basic Probability Concepts

- **Experiment:** Any process that generates observations
  - Example: Testing a patient for COVID-19
- **Sample Space (S):** Set of all possible outcomes
  - Example: $S = \{\text{Positive}, \text{Negative}\}$
- **Event:** Subset of the sample space
  - Example: Patient tests positive
- **Probability:** Measure of likelihood [0, 1]
  - $\Pr(\text{Event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$

Probability Rules

1. **Range:** $0 \leq \Pr(A) \leq 1$ for any event A
2. **Certainty:** $\Pr(S) = 1$ (something must happen)
3. **Complement:** $\Pr(A^c) = 1 - \Pr(A)$
4. **Addition Rule:**

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

5. **Multiplication Rule:**

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B \mid A)$$

Medical Example: Diagnostic Testing

COVID-19 Rapid Test Performance:

- Sensitivity $= 85\%$ (probability of positive test given disease)
- Specificity $= 97\%$ (probability of negative test given no disease)
- Disease prevalence $= 5\%$

Questions:

- What's the probability of a false positive?
- What's the probability someone with a positive test has COVID?
- How does prevalence affect test interpretation?

These questions require understanding conditional probability!

Definition: Probability of event A given that event B has occurred

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \text{ where } \Pr(B) > 0$$

Medical Example:

- $A =$ Patient has diabetes
- $B =$ Patient is overweight
- $\Pr(A \mid B) =$ Probability of diabetes given patient is overweight

Key Insight: Conditional probability updates our beliefs based on new information

Independence

Definition: Events A and B are independent if:

$$\Pr(A \mid B) = \Pr(A) \text{ or equivalently } \Pr(A \cap B) = \Pr(A) \times \Pr(B)$$

Examples in Healthcare:

- **Independent:**
  - Blood type and eye color
  - Two different patients' test results
- **Dependent:**
  - Smoking status and lung cancer
  - Age and cardiovascular disease
  - Family history and genetic disorders

Caution: Independence is often assumed but should be verified!

Diagnostic Test Performance

Key Metrics for Medical Tests:

- **Sensitivity:** Pr(Positive Test | Disease Present)
  - Ability to correctly identify disease
  - True positive rate
- **Specificity:** Pr(Negative Test | Disease Absent)
  - Ability to correctly identify no disease
  - True negative rate

Example - COVID-19 Rapid Test:

- Sensitivity = 85% (detects 85% of infected people)
- Specificity = 97% (correctly identifies 97% of non-infected)

Clinical Interpretation Measures:

- **Positive Predictive Value (PPV):** Pr(Disease | Positive Test)
  - Probability patient has disease given positive test
  - Depends on disease prevalence
- **Negative Predictive Value (NPV):** Pr(No Disease | Negative Test)
  - Probability patient is healthy given negative test
  - Also depends on disease prevalence

Key Insight: PPV and NPV change with disease prevalence in the population, while sensitivity and specificity are test characteristics.

A Counterintuitive Example

HIV Screening Test with Excellent Performance:

- Sensitivity = 99.5% (detects 995 of 1000 infected people)
- Specificity = 98.5% (correctly identifies 985 of 1000 uninfected)
- Population prevalence = 0.1% (1 in 1000 people)

Question: If you test positive, what's the probability you have HIV?

A Counterintuitive Example

HIV Screening Test with Excellent Performance:

- Sensitivity = 99.5% (detects 995 of 1000 infected people)
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Question: If you test positive, what's the probability you have HIV?

Approach: Consider 100,000 people tested

	HIV +	**HIV -**	**Total**
Test +	100	1,498	1,598
Test -	0	98,402	98,402
Total	100	99,900	100,000

PPV $= \frac{100}{1{,}598} = 0.063$ or **6.3%**

Surprising Result: Only 6.3% of positive tests indicate actual infection!
Why? With low prevalence, false positives (1,498) vastly outnumber true positives (100).

Understanding Odds

Probability vs Odds:

- **Probability:** $\Pr(\text{Event}) = \frac{\text{Number with event}}{\text{Total number}}$
- **Odds:** $\text{Odds}(\text{Event}) = \frac{\text{Number with event}}{\text{Number without event}}$

Relationship: $\text{Odds} = \frac{p}{1-p}$ where p is the probability

Example: Risk of lung cancer in smokers

- 60 smokers develop lung cancer, 440 don't
- **Risk (probability):** $\frac{60}{500} = 0.12$ (12%)
- **Odds:** $\frac{60}{440} = 0.136$ (or "60 to 440" or "3 to 22")

Key insight: When probability is small (< 0.10), odds $\approx$ probability

Relative Risk vs Odds Ratio

- **Relative Risk (RR):**
  - Direct measure of risk comparison
  - Used in cohort studies
  - RR $= 2$ means exposed have twice the risk
  - Cannot be calculated in case-control studies
- **Odds Ratio (OR):**
  - Approximates RR when disease is rare
  - Can be calculated in all study designs
  - More difficult to interpret directly
  - OR $>$ RR when outcome is common

Rule of thumb: When disease prevalence $< 10\%$, OR $\approx$ RR

Example: Smoking and Lung Cancer

Study of 1000 individuals over 20 years:

	Lung Cancer	**No Cancer**	**Total**
Smokers	60	440	500
Non-smokers	10	490	500
Total	70	930	1000

Calculations:

- Risk in smokers: $60/500 = 0.12$ (12%)
- Risk in non-smokers: $10/500 = 0.02$ (2%)
- Relative Risk: $0.12/0.02 = 6.0$
- Odds Ratio: $(60 \times 490)/(10 \times 440) = 6.68$

Interpretation: Smokers have 6 times the risk of lung cancer

Expected Value

Definition: Long-run average value of a random variable

Discrete Random Variable:

$$E(X) = \mu = \sum_i x_i \cdot \Pr(X = x_i)$$

Medical Example: Number of adverse events in clinical trial

Events (x)	0	1	2	3
Probability	0.70	0.20	0.08	0.02

$$E(X) = 0(0.70) + 1(0.20) + 2(0.08) + 3(0.02) = 0.42$$

Interpretation: On average, expect 0.42 adverse events per patient

Variance and Standard Deviation

Variance: Measure of spread around the expected value

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

Standard Deviation: Square root of variance

$$SD(X) = \sigma = \sqrt{\text{Var}(X)}$$

Properties:

- $\text{Var}(aX) = a^2\text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ if X and Y are independent
- Standard deviation has same units as original data

Binomial Distribution

Conditions:

- Fixed number of trials (n)
- Two possible outcomes (success/failure)
- Constant probability of success (p)
- Independent trials

Probability Mass Function:

$$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Parameters:

- Mean: $\mu = np$
- Variance: $\sigma^2 = np(1 - p)$

Binomial Example: Vaccine Efficacy

Clinical Trial: 20 vaccinated individuals exposed to virus

- Vaccine efficacy = 90% (probability of protection)
- What's the probability exactly 18 are protected?

Solution:

- $n = 20$, $p = 0.90$, $k = 18$
- $\Pr(X = 18) = \binom{20}{18}(0.90)^{18}(0.10)^2$
- $\Pr(X = 18) = 190 \times 0.1501 \times 0.01 = 0.285$

Expected number protected: $\mu = 20 \times 0.90 = 18$
Standard deviation: $\sigma = \sqrt{20 \times 0.90 \times 0.10} = 1.34$

Normal Distribution

Properties:

- Bell-shaped, symmetric curve
- Characterized by mean (μ) and standard deviation (σ)
- Area under curve $= 1$
- Approximately 68% within $\pm 1\sigma$
- Approximately 95% within $\pm 2\sigma$
- Approximately 99.7% within $\pm 3\sigma$

Standard Normal: $Z = \frac{X-\mu}{\sigma}$ has mean 0, SD 1

Medical Applications:

- Blood pressure measurements
- Cholesterol levels
- Height and weight
- Many lab values

Normal Distribution Example

Systolic Blood Pressure in Adults:

- Mean $= 120$ mmHg
- Standard deviation $= 15$ mmHg
- What proportion have BP > 140 mmHg (hypertension)?

Solution:

1. Standardize: $Z = \frac{140-120}{15} = 1.33$
2. Look up in Z-table: $\Pr(Z < 1.33) = 0.9082$
3. Therefore: $\Pr(BP > 140) = 1 - 0.9082 = 0.0918$

Interpretation: About 9.2% of adults have hypertension by this criterion

Probability in Public Health

Disease Surveillance Example:

- Population: 50,000 individuals
- New disease cases: 125 per year
- Deaths from disease: 15 per year

Calculate probabilities:

- $\Pr(\text{disease}) = 125/50{,}000 = 0.0025\ (0.25\%)$
- $\Pr(\text{death} \mid \text{disease}) = 15/125 = 0.12\ (12\%)$
- $\Pr(\text{death and disease}) = 15/50{,}000 = 0.0003\ (0.03\%)$

Clinical interpretation:

- Incidence rate: 250 per 100,000 per year
- Case fatality rate: 12%
- Overall mortality rate: 30 per 100,000 per year

AI for Probability Calculations

How AI can help with probability:

- **Complex calculations:** Conditional probabilities, combinatorics
- **Simulation:** Generate random data, bootstrap samples
- **Visualization:** Create probability distributions
- **Problem solving:** Step-by-step solutions
- **Code generation:** R scripts for probability problems

Example prompts:

- "Calculate the positive predictive value for a test with 95% sensitivity, 90% specificity, and 5% prevalence"
- "Simulate 10,000 binomial trials with n=20, p=0.3 in R"
- "Explain the difference between relative risk and odds ratio with a medical example"

AI Demo: Diagnostic Test Calculations

Prompt: "Create R code to calculate diagnostic test metrics"

```
# Diagnostic Test Calculator
diagnostic_metrics <- function(sens, spec, prev) {
  # Calculate PPV and NPV
  ppv <- (sens * prev) /
         (sens * prev + (1-spec) * (1-prev))
  npv <- (spec * (1-prev)) /
         ((1-sens) * prev + spec * (1-prev))

  # Display results
  cat("PPV:", round(ppv*100, 1), "%\n")
  cat("NPV:", round(npv*100, 1), "%\n")

  return(list(PPV=ppv, NPV=npv))
}

# Example: COVID test
diagnostic_metrics(sens=0.85, spec=0.97, prev=0.05)
```

AI Demo: Probability Simulations

Prompt: "Simulate coin flips to estimate probability"

```
# Simulate binomial trials
simulate_treatment <- function(n_patients=100,
                               p_success=0.7) {
  outcomes <- rbinom(n_patients, 1, p_success)
  success_rate <- mean(outcomes)

  cat("Simulated success rate:",
      round(success_rate*100, 1), "%\n")
  cat("Expected success rate:", p_success*100, "%\n")

  # Create visualization
  barplot(table(outcomes),
          names.arg=c("Failure", "Success"),
          col=c("red", "green"),
          main="Treatment Outcomes")
}

simulate_treatment()
```

AI Demo: Visualizing Distributions

Prompt: “Compare binomial and normal distributions in R”

```
# Compare Binomial to Normal
n <- 100; p <- 0.3
x <- 0:n

# Binomial probabilities
binom_probs <- dbinom(x, n, p)

# Normal approximation
mu <- n * p
sigma <- sqrt(n * p * (1-p))

# Plot comparison
plot(x, binom_probs, type="h", col="blue",
     main="Binomial vs Normal",
     xlab="x", ylab="Probability")
curve(dnorm(x, mu, sigma), add=TRUE, col="red")
legend("topright", c("Binomial", "Normal"),
       col=c("blue", "red"), lty=1)
```

AI Best Practices for Probability

DO:

- Verify calculations manually
- Check assumptions are met
- Request step-by-step solutions
- Ask for multiple approaches
- Validate with simulations
- Cross-reference formulas

DON'T:

- Accept complex calculations blindly
- Skip understanding the logic
- Ignore edge cases
- Misinterpret conditional probability
- Confuse probability types
- Trust without verification

Remember: AI can calculate quickly, but you need to understand the concepts!

AI Task: Probability Problem Solving

This week's AI-integrated tasks:

1. Use AI to solve a diagnostic test problem, then verify manually
2. Generate R code for simulating coin flips, then modify for medical scenario
3. Ask AI to explain conditional probability three different ways
4. Create visualizations of different probability distributions
5. Identify and correct an AI error in probability calculation

Sample Problem:

```
''A new diagnostic test for Disease X has 92% sensitivity
and 88% specificity. In a population where 3% have the
disease, what is the probability that someone with a
positive test actually has the disease? Create R code
to calculate this and visualize how the answer changes
as prevalence varies from 1% to 10%.''
```

R Example: Basic Probability

```
# Probability calculations in R
# Sample space for rolling two dice
dice1 <- 1:6
dice2 <- 1:6
sample_space <- expand.grid(dice1, dice2)
sample_space$sum <- sample_space$Var1 + sample_space$Var2

# Probability of sum = 7
p_seven <- sum(sample_space$sum == 7) / nrow(sample_space)
cat("Pr(sum = 7) =", p_seven, "\n")  # 6/36 = 1/6

# Conditional probability example
# Pr(sum > 8 | first die = 4)
subset_data <- sample_space[sample_space$Var1 == 4, ]
p_conditional <- sum(subset_data$sum > 8) / nrow(subset_data)
cat("Pr(sum > 8 | first = 4) =", p_conditional, "\n")  # 3/6 = 1/2
```

R Example: Medical Testing

```r
# Breast cancer screening example
sens <- 0.90  # Sensitivity
spec <- 0.95  # Specificity
prev <- 0.008 # Prevalence

# Calculate PPV and NPV
ppv <- (sens * prev) /
       (sens * prev + (1-spec) * (1-prev))
npv <- (spec * (1-prev)) /
       ((1-sens) * prev + spec * (1-prev))

cat("PPV:", round(ppv * 100, 1), "%\n")
cat("NPV:", round(npv * 100, 1), "%\n")

# Simulate population
n <- 10000
n_disease <- round(n * prev)
n_healthy <- n - n_disease

# Test outcomes
TP <- round(n_disease * sens)
```

R Example: Binomial Distribution

```
# Clinical trial with 20 patients, treatment success rate 70%
n <- 20
p <- 0.7

# Probability of exactly 15 successes
p_15 <- dbinom(15, n, p)
cat("Pr(X = 15) =", round(p_15, 4), "\n")

# Probability of at least 15 successes
p_atleast_15 <- pbinom(14, n, p, lower.tail = FALSE)
cat("Pr(X >= 15) =", round(p_atleast_15, 4), "\n")

# Expected value and standard deviation
expected <- n * p
std_dev <- sqrt(n * p * (1-p))
cat("Expected successes:", expected, "\n")
cat("Standard deviation:", round(std_dev, 2), "\n")

# Visualize distribution
x <- 0:n
probs <- dbinom(x, n, p)
```

R Example: Normal Distribution

```
# Blood pressure in adults
mu <- 120       # Mean SBP
sigma <- 15     # Standard deviation

# Proportion with hypertension (SBP > 140)
p_hyper <- pnorm(140, mu, sigma, lower.tail = FALSE)
cat("Pr(BP > 140):", round(p_hyper * 100, 1), "%\n")

# Find percentiles
bp_90 <- qnorm(0.90, mu, sigma)
bp_95 <- qnorm(0.95, mu, sigma)
cat("90th percentile:", round(bp_90, 1), "mmHg\n")
cat("95th percentile:", round(bp_95, 1), "mmHg\n")

# Visualize distribution
x <- seq(70, 170, length = 100)
plot(x, dnorm(x, mu, sigma), type = "l",
     main = "Systolic Blood Pressure",
     xlab = "SBP (mmHg)", ylab = "Density")
abline(v = c(mu, 140), col = c("blue", "red"))
```

Summary: Chapter 6 - Probability

Today we covered:

- **Section 6.1:** Basic probability concepts, rules, and definitions
- **Section 6.2:** Conditional probability and independence
- **Section 6.3:** Diagnostic test applications (sensitivity, specificity)
- **Section 6.4:** Random variables and expected value
- **Section 6.5:** Discrete distributions (binomial)

Key Learning Objectives:

- Calculate and interpret basic probabilities
- Understand conditional probability and independence
- Apply probability concepts to diagnostic testing
- Use probability distributions for modeling

Next Week: Chapter 7 & 8 - Theoretical distributions and sampling

Questions?

Thank you!

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Reading for next week:
Pagano & Gauvreau, Chapter 7 (7.1, 7.2, 7.4) and Chapter 8